

Ajiteshwar Singh Sandhu

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EDUCATION

Thapar Institute of Engineering & Technology, Patiala, Punjab

2027

B.E. in Electronics & Communication Engineering | CGPA: 9.04/10

Budha Dal Public School, Patiala, Punjab

2023

Class XII, Non-Medical | 91.2% | Secured 3rd position in Non-Medical stream

EXPERIENCE

DSAI, Thapar Institute of Engineering & Technology, Patiala

Sep 2025 – May 2026

Research Intern

- Designed **MambaWeed**, a YOLOv8m-based architecture integrating four-directional Mamba State Space Model (SSM) blocks for fine-grained 14-class crop and weed detection.
- Trained and evaluated the model on a unified agricultural dataset, achieving **89.9% Precision, 90.2% Recall, 93.5% mAP@0.5, and 86.5% mAP@0.5:0.95**.
- Developed a custom **MambaTrainer** to resolve Ultralytics checkpoint serialization issues, enabling reliable training and persistence of custom SSM-augmented model weights.
- Achieved **10.9 ms inference time** per image on an NVIDIA RTX 4050 while maintaining 79.1 GFLOPs and introducing only 579 additional trainable parameters over the YOLOv8m baseline.
- Co-authored the corresponding research paper, accepted and published in the **NGNDAI 2026** conference proceedings, and opensourced the implementation on GitHub. GitHub: [MambaWeed](#)

PROJECTS

AI-Governed Blockchain-Based Carbon Credit Validation Framework

React.js, TypeScript, FastAPI, PostgreSQL, XGBoost, Gemini API, Solidity, Web3.py, Polygon, ERC-1155

- Collaborated in a 5-member team to develop a full-stack carbon credit validation platform integrating React, FastAPI, PostgreSQL, AI-based risk assessment, and blockchain smart contracts.
- Led blockchain development, designing and deploying Solidity smart contracts on Polygon for multi-validator consensus, ERC1155 carbon credit issuance, unique credit identity tracking, and retirement enforcement.
- Integrated deployed smart contracts with the FastAPI backend using Web3.py, enabling end-to-end project submission, validation, governance, and token issuance.
- Co-authored a research paper on the proposed framework, submitted to **IEEE DELCON 2026** and currently under review. GitHub: [AI-Governed-Carbon-Credit-Framework](#)

IoT-Based Overspeed Detection & Alert System

ESP32, NEO-6M GPS, Ultrasonic Sensor, Arduino, TinyGPS++, Embedded C/C++

- Designed and implemented an IoT-based vehicle overspeed detection system using ESP32, NEO-6M GPS, and ultrasonic sensing to monitor real-time vehicle speed and trigger automated alerts upon threshold violations.
- Integrated GPS-based positioning with distance-sensor measurements to cross-validate speed estimation and improve detection reliability over a single-sensor approach.
- Engineered an end-to-end embedded pipeline encompassing sensor data acquisition, real-time speed computation, threshold-based detection, and alert triggering, while resolving hardware-level challenges involving UART communication and GPS satellite-fix reliability.

TECHNICAL SKILLS

Programming Languages: C++, Python, TypeScript, SQL, Solidity

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks

Embedded & Hardware: Embedded C/C++, ESP32, Arduino, Microcontrollers, UART, I2C, SPI, GPIO, PWM, ADC, Sensor Interfacing, GPS, IoT, Digital Electronics, Microprocessors & Microcontrollers

Frameworks & Libraries: React.js, FastAPI, PyTorch, XGBoost, OpenCV

Databases: PostgreSQL, MySQL

Blockchain: Solidity, Ethereum, Web3.py, ERC-1155, Smart Contracts, Polygon **Developer**

Tools: Git, GitHub, Linux, Postman, Jupyter Notebook, LaTeX

RELEVANT COURSEWORK

Data Structures & Algorithms | Object-Oriented Programming | Database Management Systems | Computer Networks | Digital Electronics | Microprocessors & Microcontrollers | Computer Architecture | Embedded Systems | Machine Learning

ACHIEVEMENTS & PUBLICATIONS

- Co-authored *MambaWeed: Fine-Grained Multi-Species Weed Detection Using Four-Directional Mamba Blocks in YOLOv8 for Precision Agriculture*, accepted and published in the NGNDAI 2026 conference proceedings.
- Co-authored research on an *AI-Governed Blockchain-Based Carbon Credit Validation Framework*, submitted to IEEE DELCON 2026 (under review).
- Recipient of the **Meritorious Honour Award** for outstanding academic performance in Classes X and XII.